Sets 1 — Set notation · Venn diagrams

Notation: $\in$ is a member of · $\notin$ is not a member of · $n(A)$ number of elements · $\varnothing$ empty set · $\mathscr{E}$ universal set · A' complement · $A \cap B$ intersection · $A \cup B$ union · $B \subset A$ subset. Marks in brackets. ★ = harder question.

1 Say whether each statement is true or false. [4]

a $7 \in \{\text{prime numbers}\}$

b $15 \notin \{\text{multiples of } 3\}$

c square $\in \{\text{quadrilaterals}\}$

d $0.5 \in \{\text{integers}\}$

2 $\mathscr{E} = \{\text{integers from 1 to 12}\}$, $A = \{\text{even numbers}\}$, $B = \{\text{multiples of } 3\}$. [6]

a List the set $A \cap B$.

b List the set $A \cup B$.

c Find $n(B')$.

d Describe the set $A \cap B$ in words.

3 In a class of 30 students, $F = \{\text{plays football}\}$ and $T = \{\text{plays tennis}\}$. 12 students play football only, 5 play both, and 8 play tennis only. [5]

a Draw a Venn diagram showing this information.

b How many students play neither sport?

c Find $n(F \cup T)$.

d Find $n(T)$.

4 $\mathscr{E} = \{\text{cars in a car park}\}$, $R = \{\text{red cars}\}$, $E = \{\text{electric cars}\}$. Describe in words: [4]

a the set R'

b the set $R \cap E$

c the set $R \cup E$

d what it means if $R \cap E = \varnothing$.

5 For two sets P and Q: $n(\mathscr{E}) = 40$, $n(P) = 22$, $n(Q) = 19$ and $n(P \cap Q) = 7$. Find $n(P \cup Q)$ and $n\big((P \cup Q)'\big)$. [3]

6 $\mathscr{E} = \{\text{positive integers up to } 20\}$, $M = \{\text{multiples of } 4\}$, $E = \{\text{even numbers}\}$. [4]

a List M and explain why $M \subset E$.

b Find $n(E \cap M')$ and describe this set.

Total: 26 marks

Check: $\cap$ = "and" (overlap), $\cup$ = "or" (everything inside), $'$ = "not" · don't double-count the overlap · the number outside the circles is part of $\mathscr{E}$ too.